



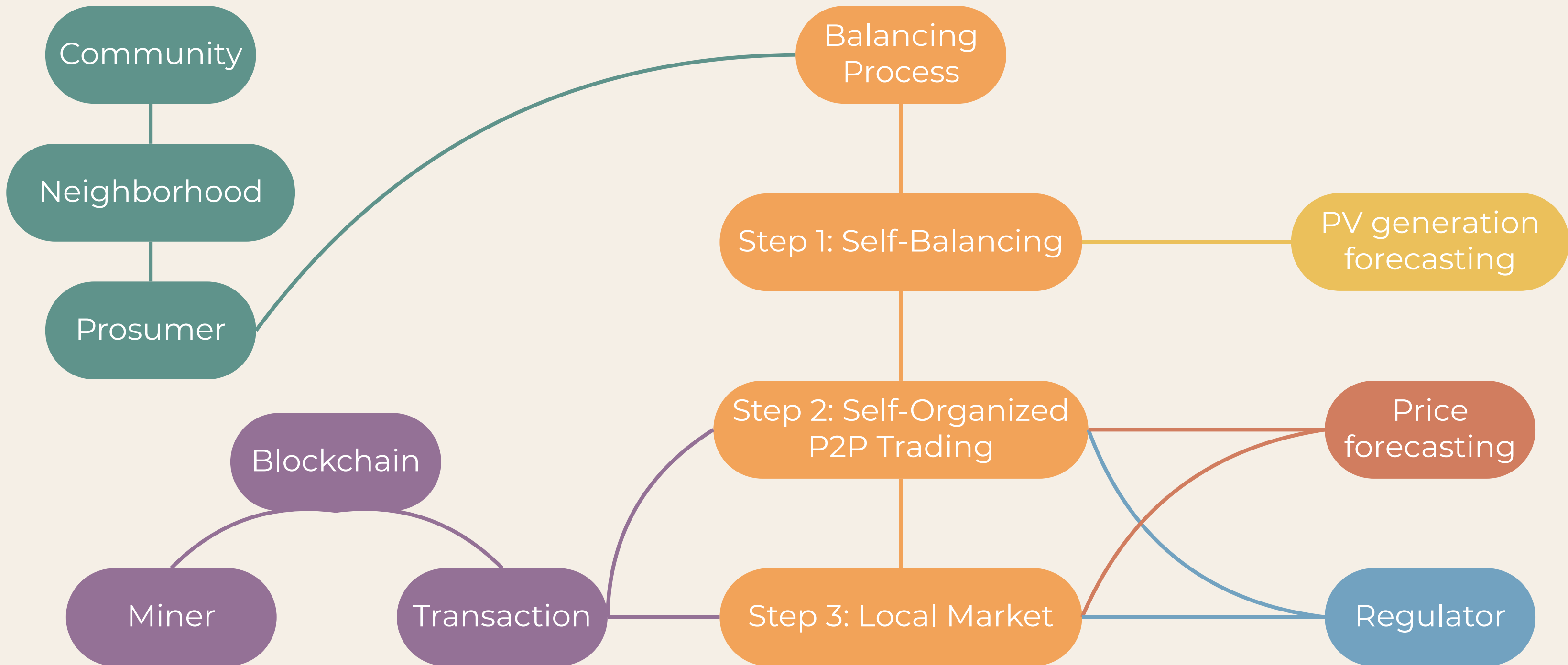
**Politecnico
di Torino**

PROSUMERS COMMUNITY SIMULATOR

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System Architecture



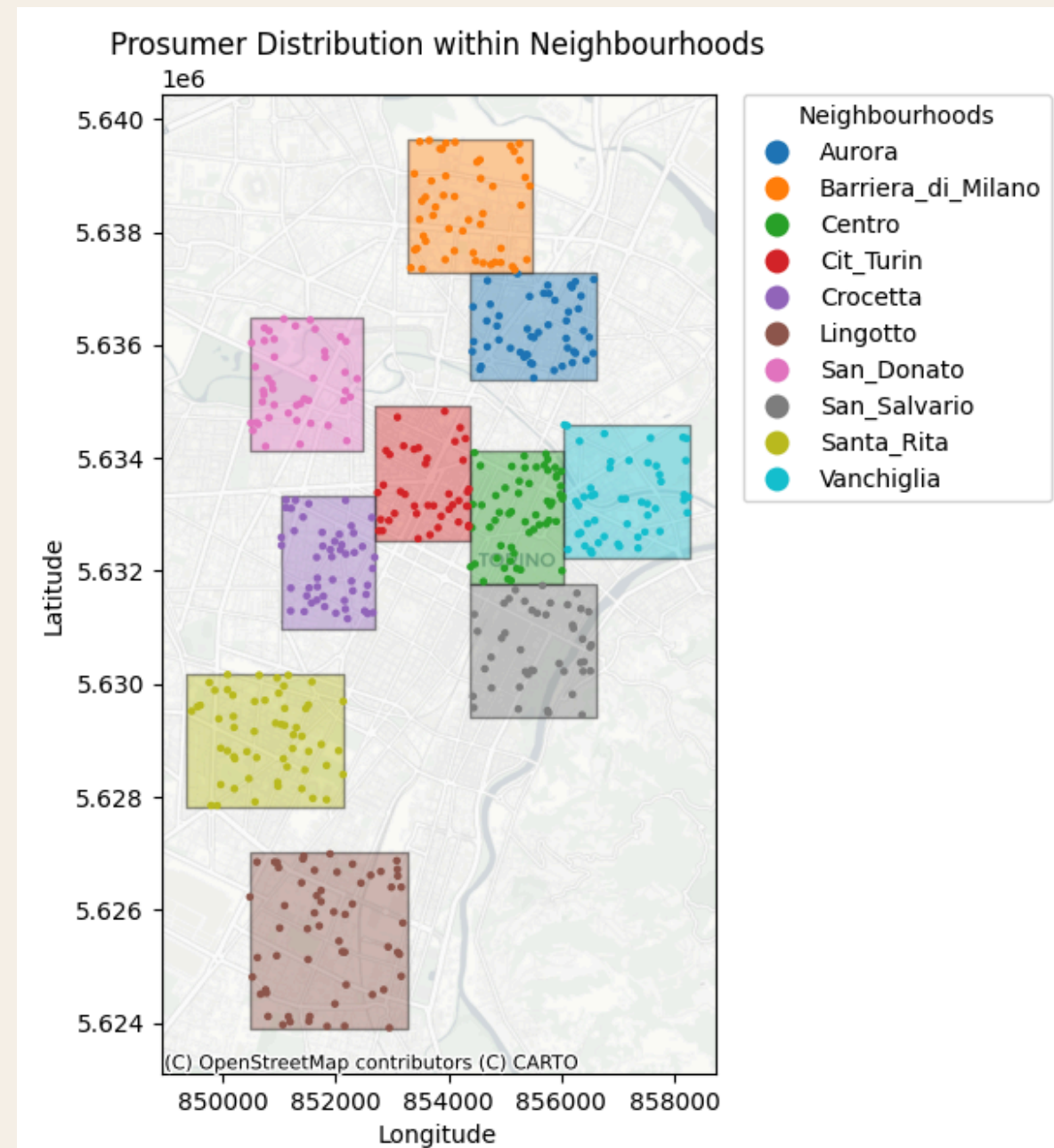
Community

Neighbourhoods: predefined areas in Turin with a given name and boundaries

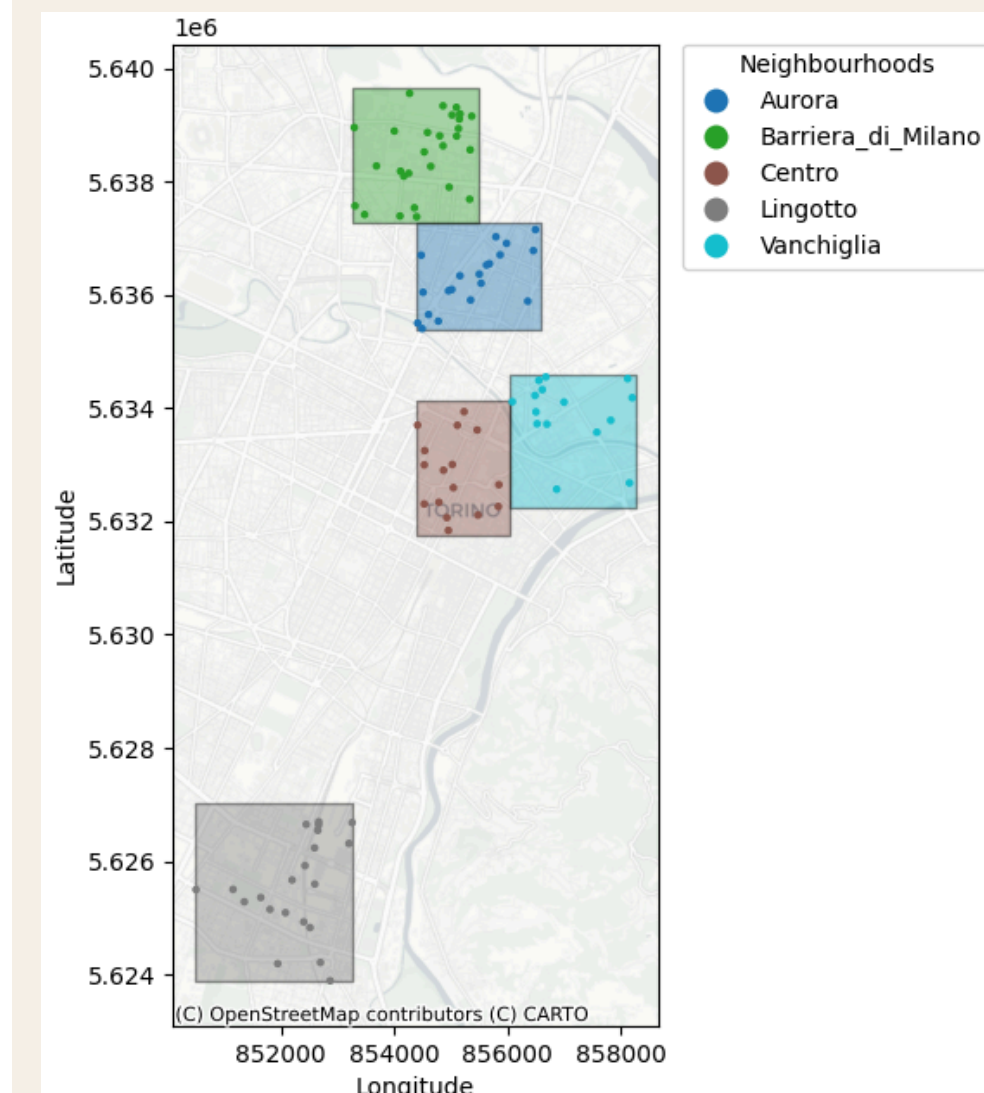
Prosumers: generate and consume energy for their needs, are equipped with a battery

Parameters:

- PV capacity: 0-20 panels $\times$ 0.25 kW each
- Base load: 0.5, 1.5 kW ($\times$ load profile)
- Battery capacity: 0, 10, or 15 kWh
- Neighbourhood: randomly selected
- Coordinates: randomly sampled inside the neighbourhood



PV generation model training:
500 prosumers - 10 neighbourhoods



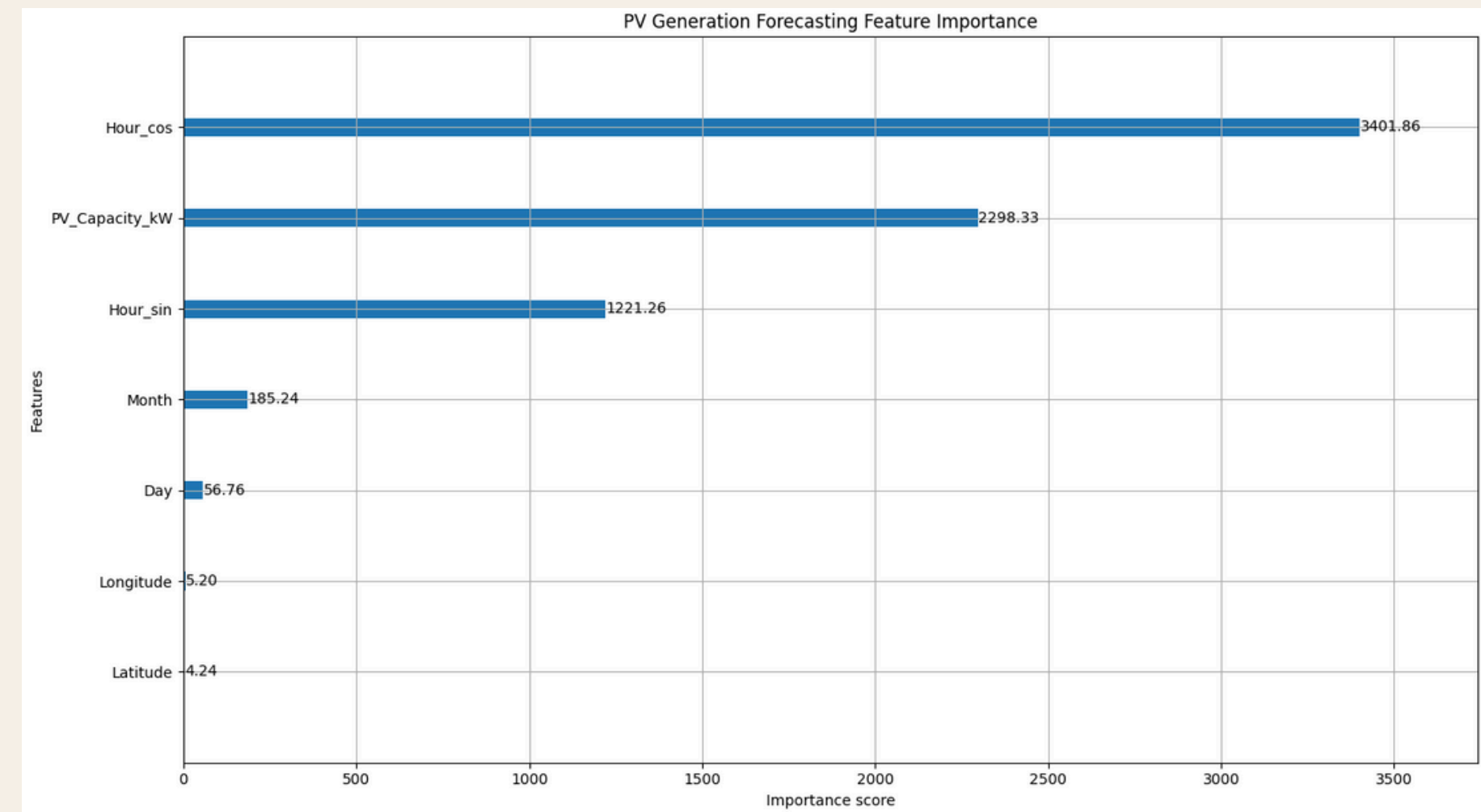
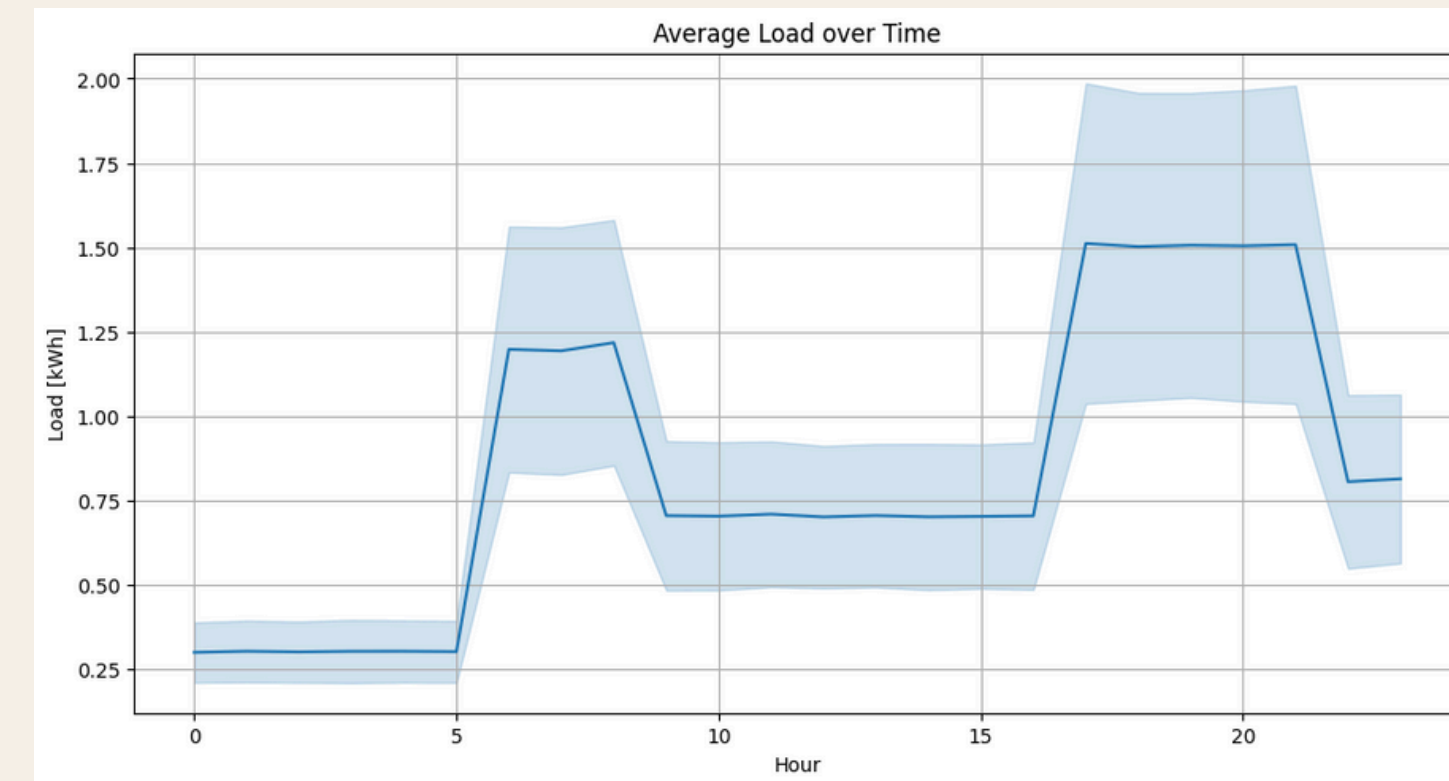
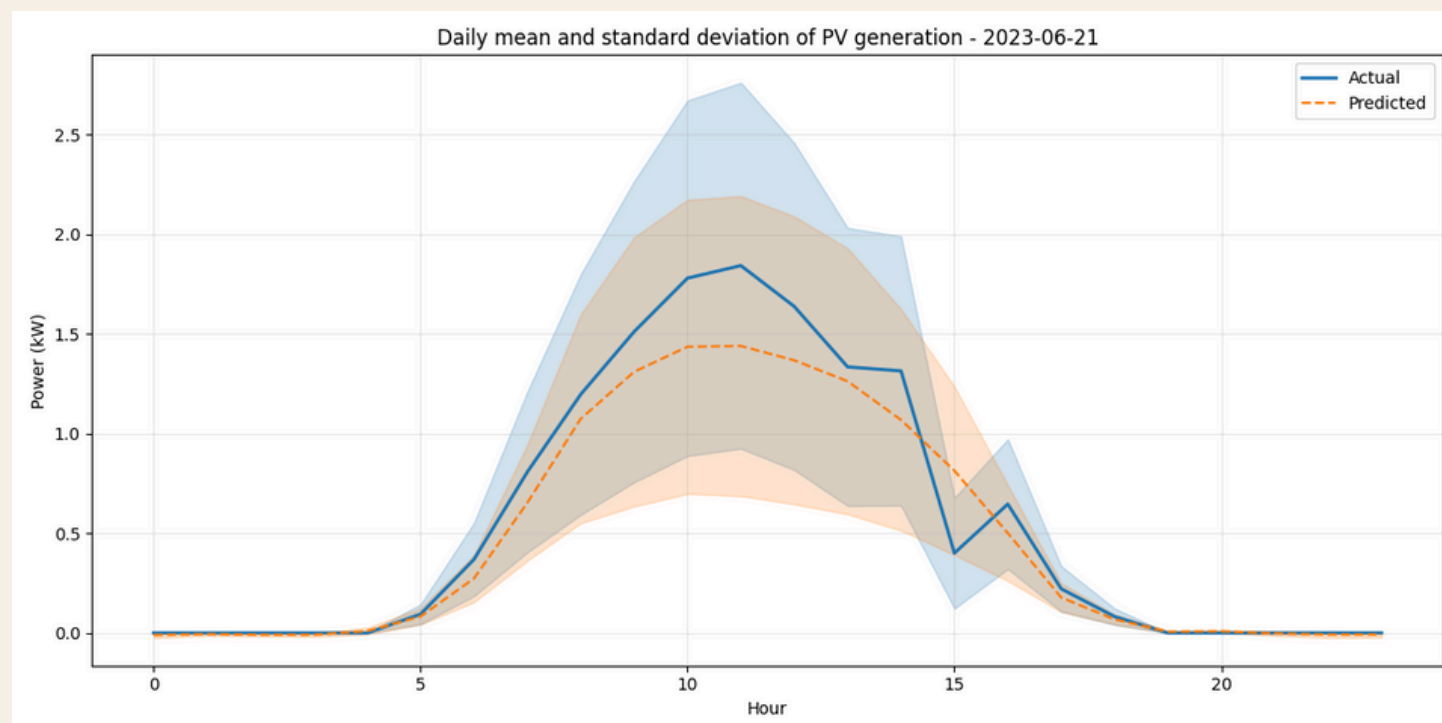
Simulation:
100 prosumers - 5 neighbourhoods

Prosumers

Load profile: 24-hour energy load profile × base load

PV generation: model predicts the generation of the PV panels of each prosumer at each hour

- Historical data: Hourly PV generation from PVGIS for 500 prosumers (2021-2023)
- Model: XGBoost regressor*
 - Tuned with Cross-Validation
 - Trained on 80% train - 20% test
- **Results:** RMSE=0.350 kW, $R^2=0.8095$



* XGBoost achieved highest accuracy for very short-term PV forecasting (reference [link](#))

Regulator

Objective: Maximize P2P Trading

How? Dynamic Incentive System

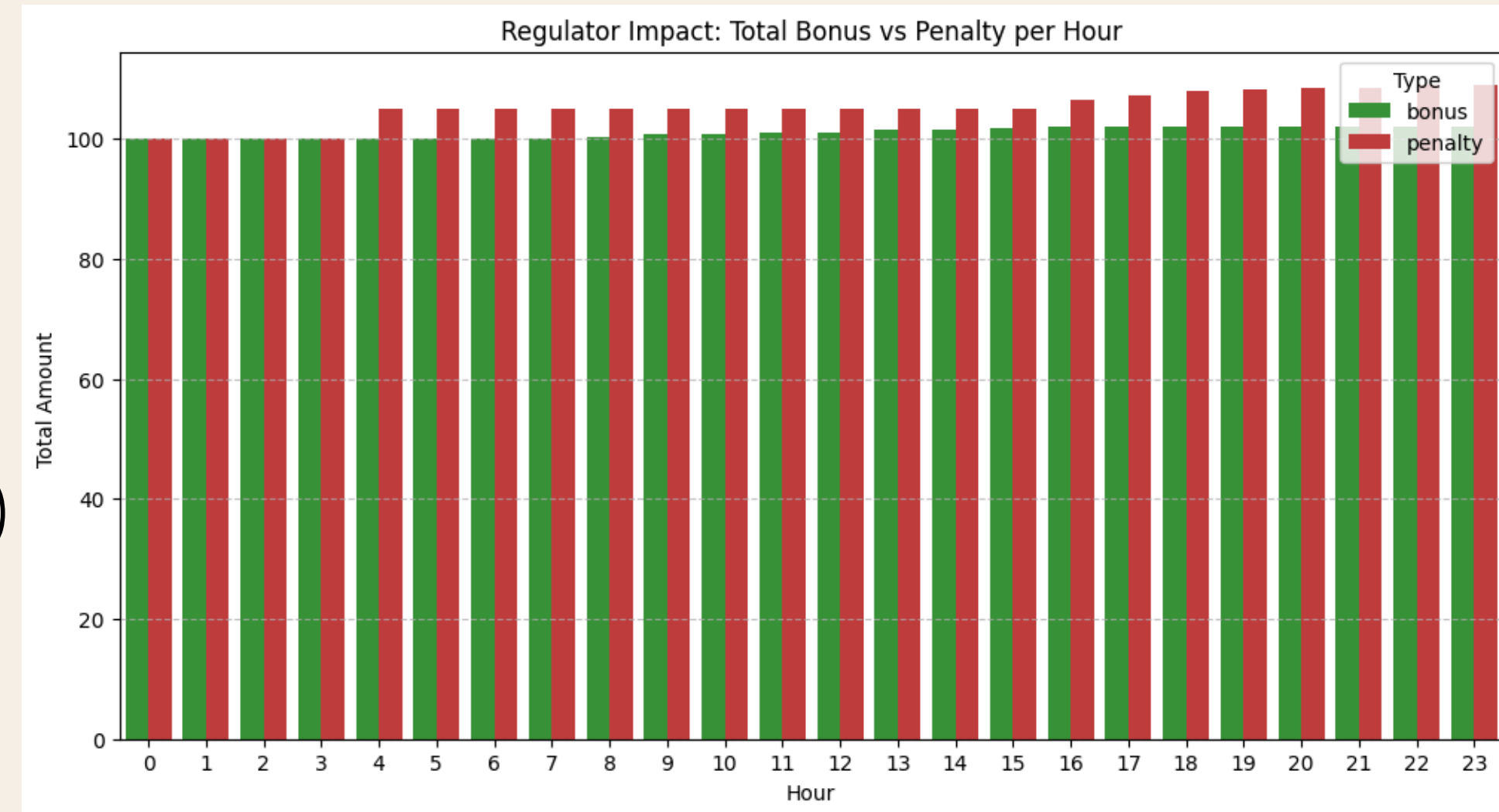
Reward P2P participation

- 1+ P2P trades → 1.02× (2% advantage)
- 5+ P2P trades → 1.05× (5% advantage)
- 10+ P2P trades → 1.10× (10% advantage)
- Buyers pay less, sellers earn more in P2P market

Discourage grid dependency

- 5+ grid purchases → 1.05× (5% higher cost)
- 10+ grid purchases → 1.10× (10% higher cost)
- Progressive price increase for grid reliance

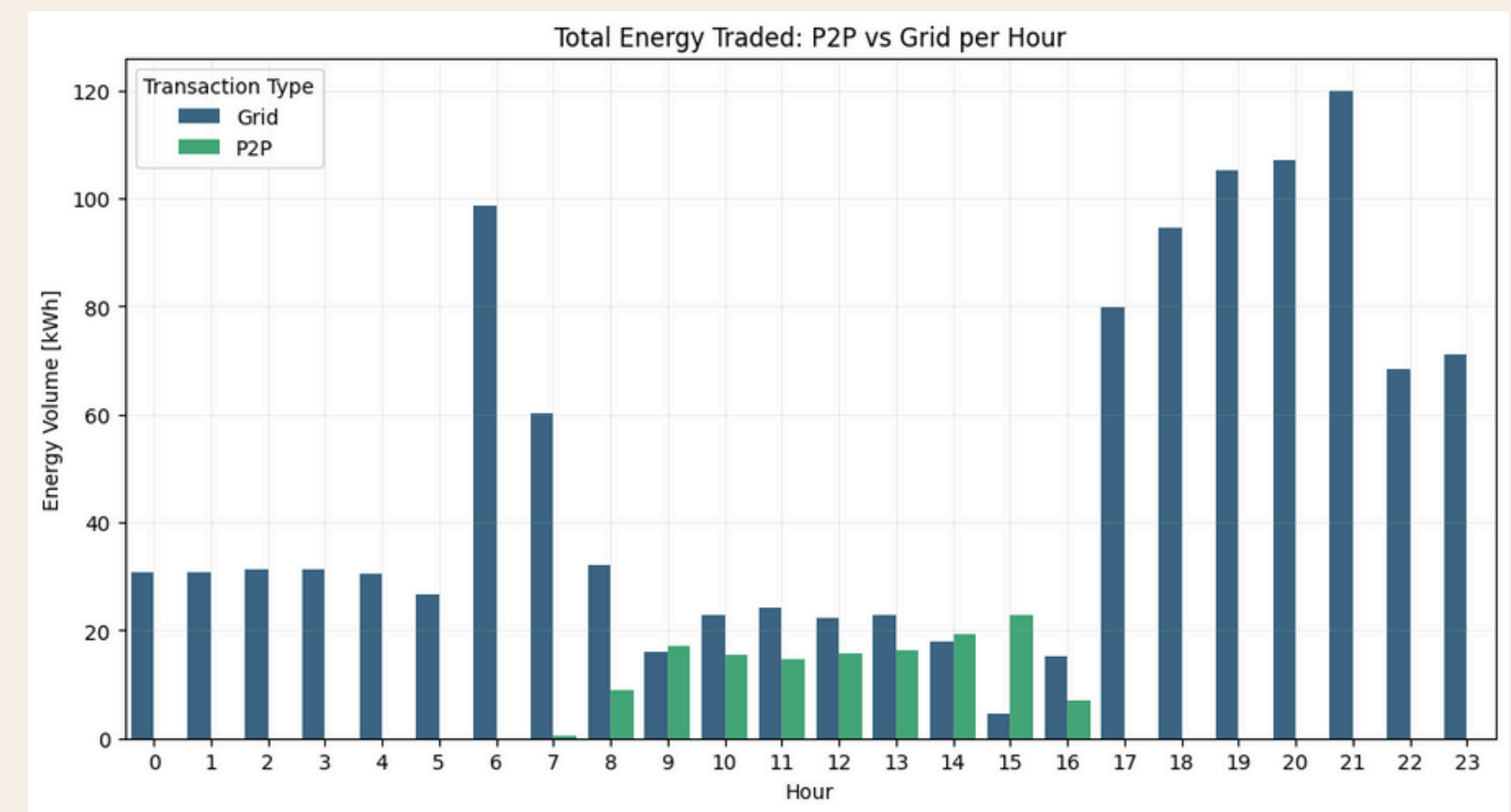
After each hour, counters updated → multipliers recalculated



Balancing Process

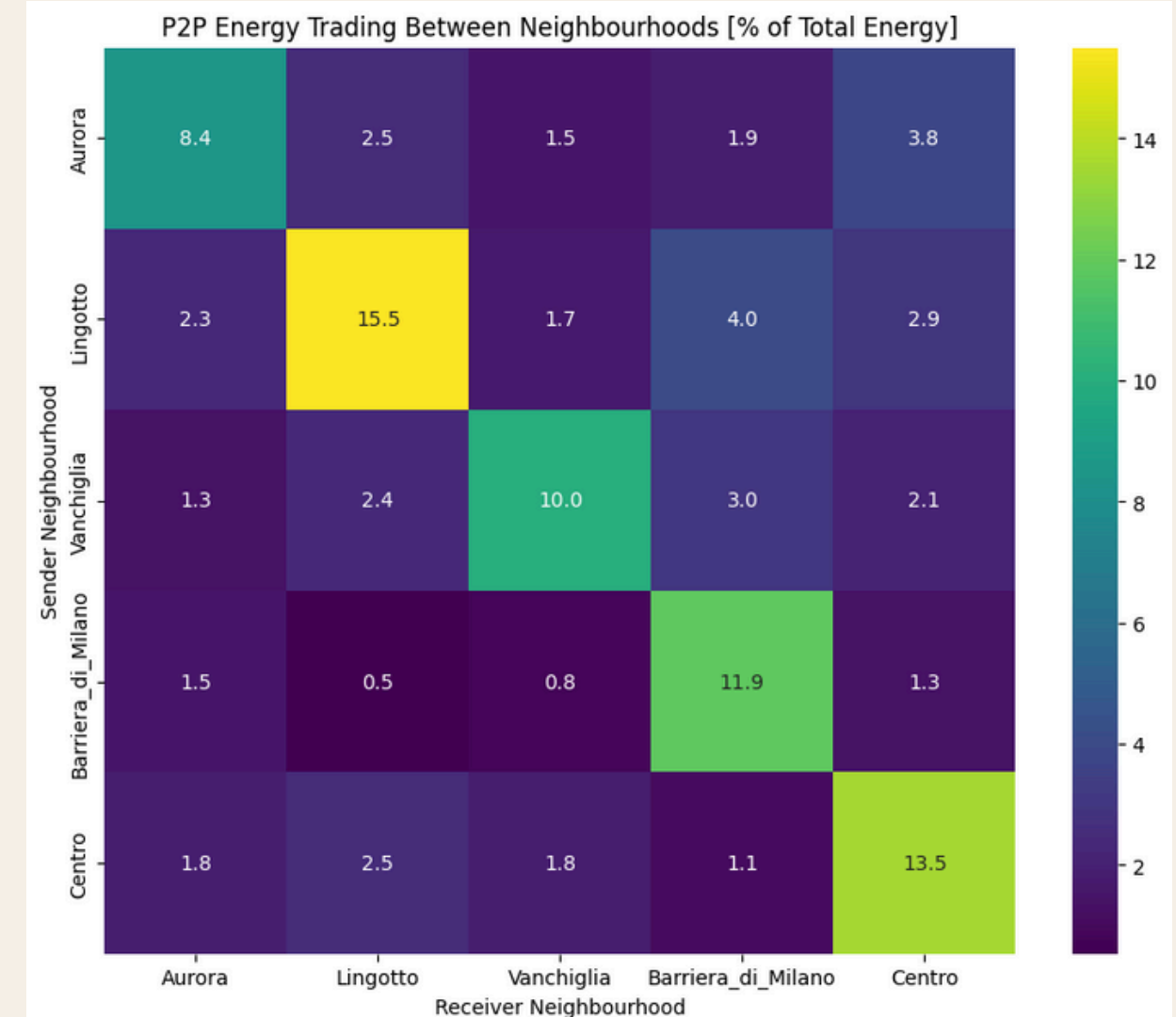
Step 1: Self-Balancing

- Calculate imbalance = load - pv_generation
- Battery usage:
 - Deficit → discharge up to available level
 - Surplus → charge up to capacity limit



Step 2: Self-Organized P2P Trading

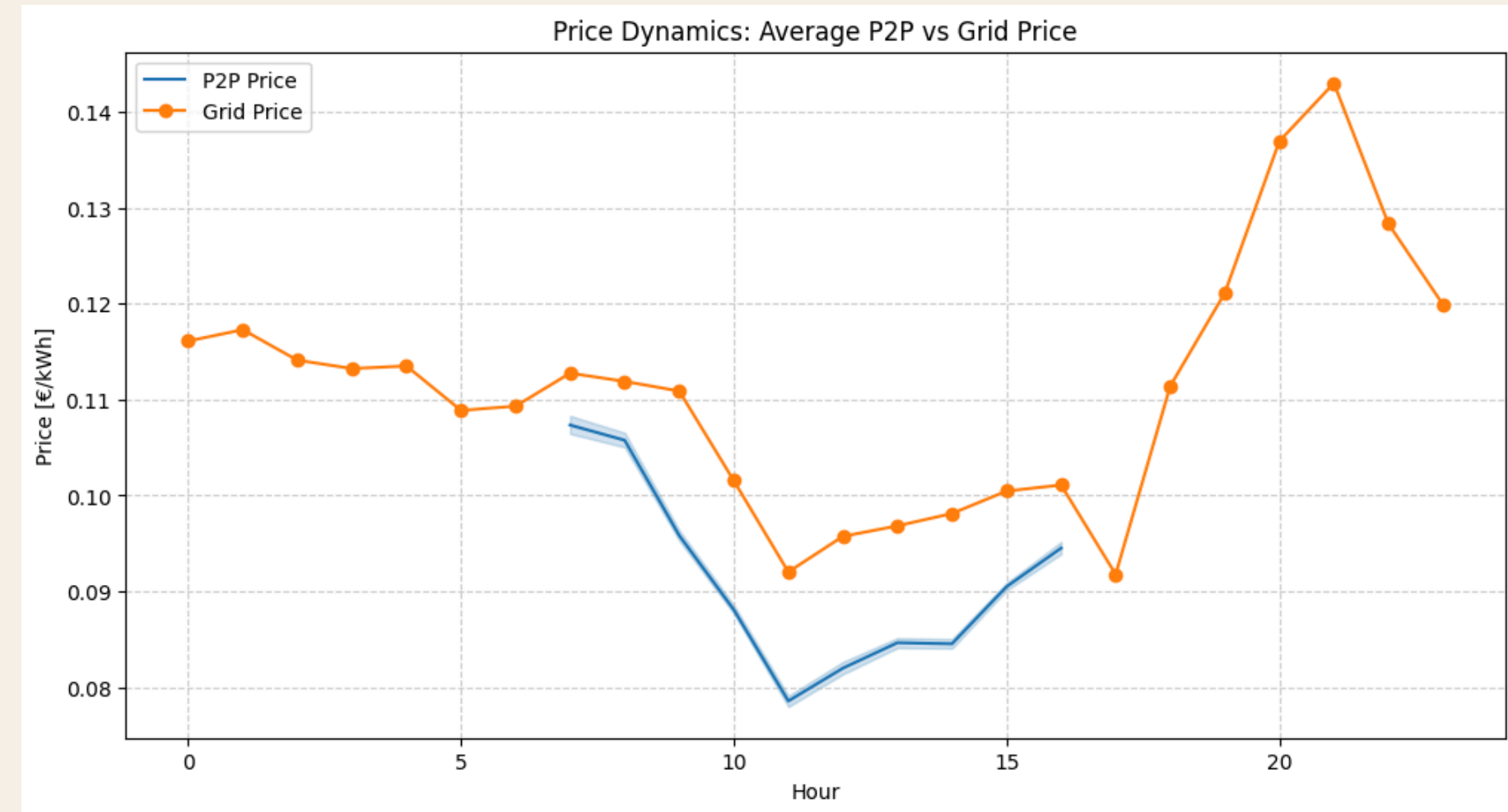
- Sorting strategy:
 - Sellers: cheapest first (sell at 70-90% market price)
 - Buyers: highest bid first (buy at 90-110% market price)
- Trade when: $\text{buyer_bid} \geq \text{seller_ask}$
- Transaction price = average of bid & ask
- P2P reward factors of regulator applied
- Network fees: 2% for cross-neighbourhood trades



Balancing Process

Step 3: Local Market (Grid/Aggregator)

- Clearing remaining imbalances:
 - Buying from grid:
 $\text{price} = \text{market_price} \times (1 + 5\% \text{ fee}) \times \text{penalty_factor}$
 - Selling to grid:
 $\text{price} = \text{market_price} \times (1 - 5\% \text{ fee})$
- Aggregator fee: 5% margin on all grid trades
- Grid penalty factors of regulator applied



→ Progressive cost increase drives prosumers towards P2P trading and battery usage

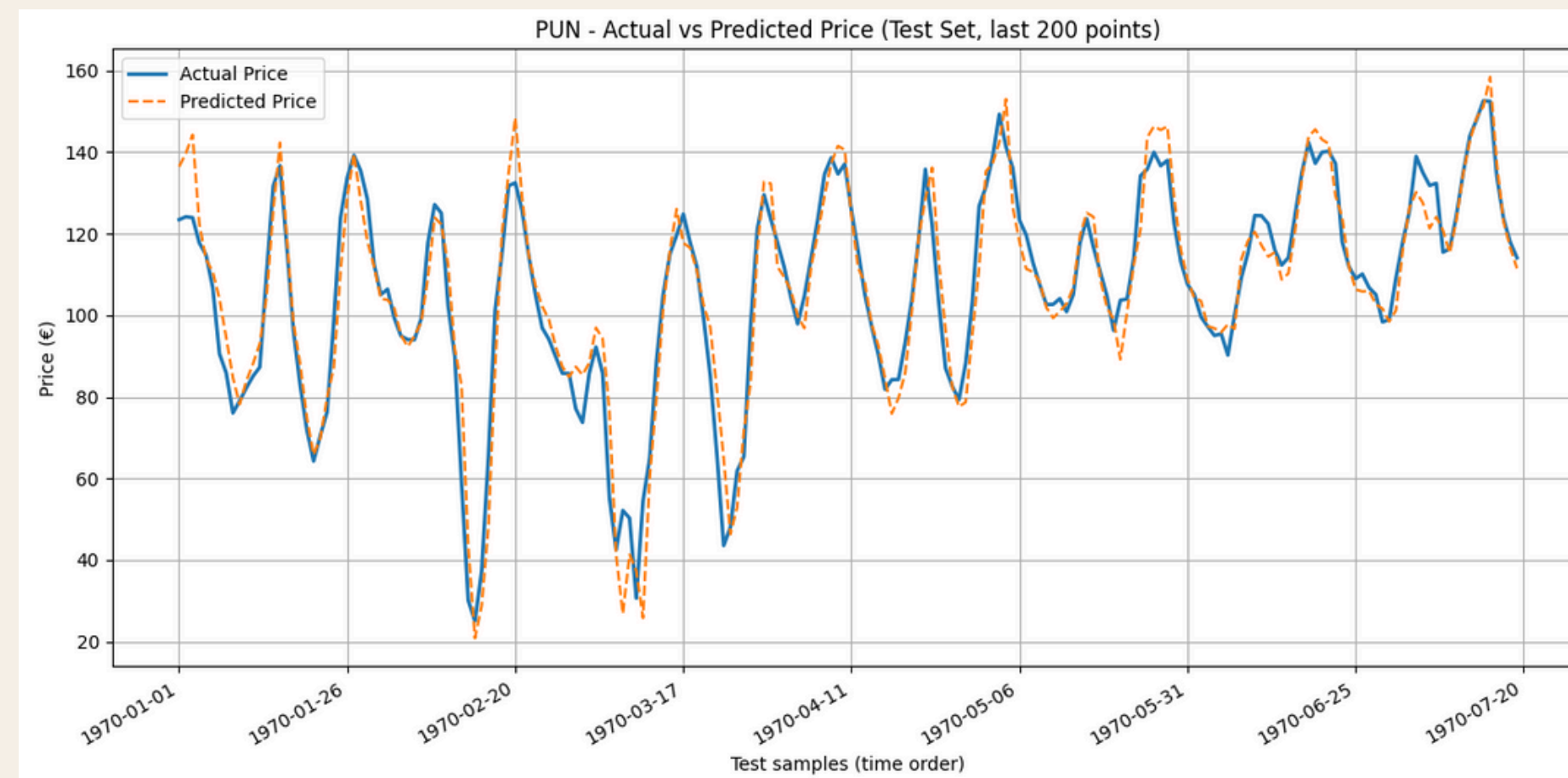
Price Forecasting

Objective: Provide a reference price for prosumer trading decisions

How? Forecasts next-hour electricity prices across market zones at each hour

- Historical data: Italian electricity market data (2021-2025)
- Model: Multi-output LightGBM regressor*
 - Trained on chronological split: 80% train - 20% test

Results: PUN: RMSE=7.751 €/MWh, $R^2=0.941$



* LightGBM achieved highest accuracy across 3 European markets with 45-60 day windows (reference [link](#))

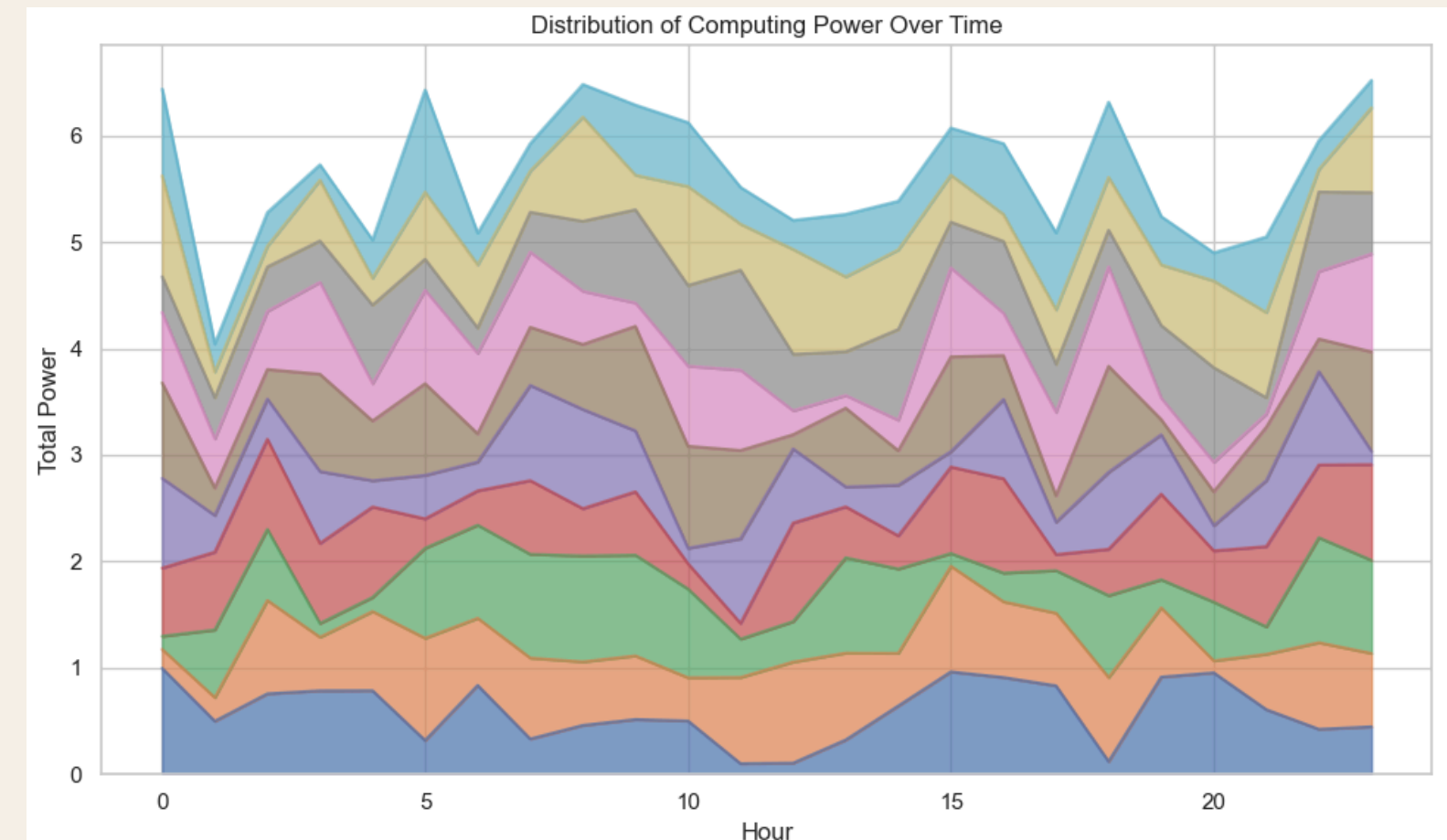
Blockchain

Transaction: Every energy trade creates a transaction containing:

- Sender and receiver IDs
- Exchanged energy amount (kWh)
- Price per kWh (€/kWh)
- Simulation step (hour 0 - 23)
- Timestamp of the transaction

Proof-of-Work mining:

- Miners: 10 independent mining nodes with random hash power (0.1-1.0)
- Difficulty: blocks must start with 3 leading zeros
- Winner selection: probabilistic based on hash power (to simulate PoW competition)
- Block creation: winner packages pending transactions and mines new block
- Validation: SHA-256 hashing with nonce iteration until target is met

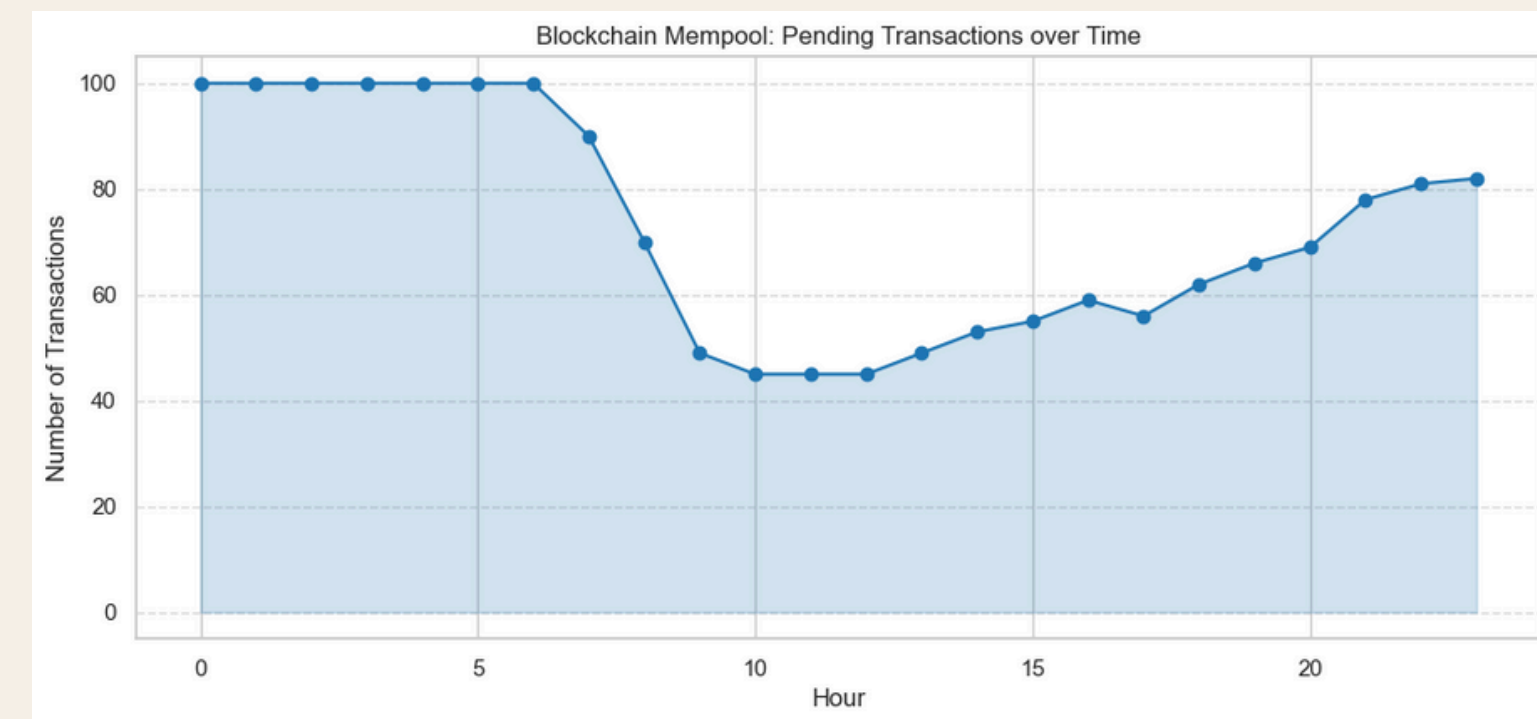
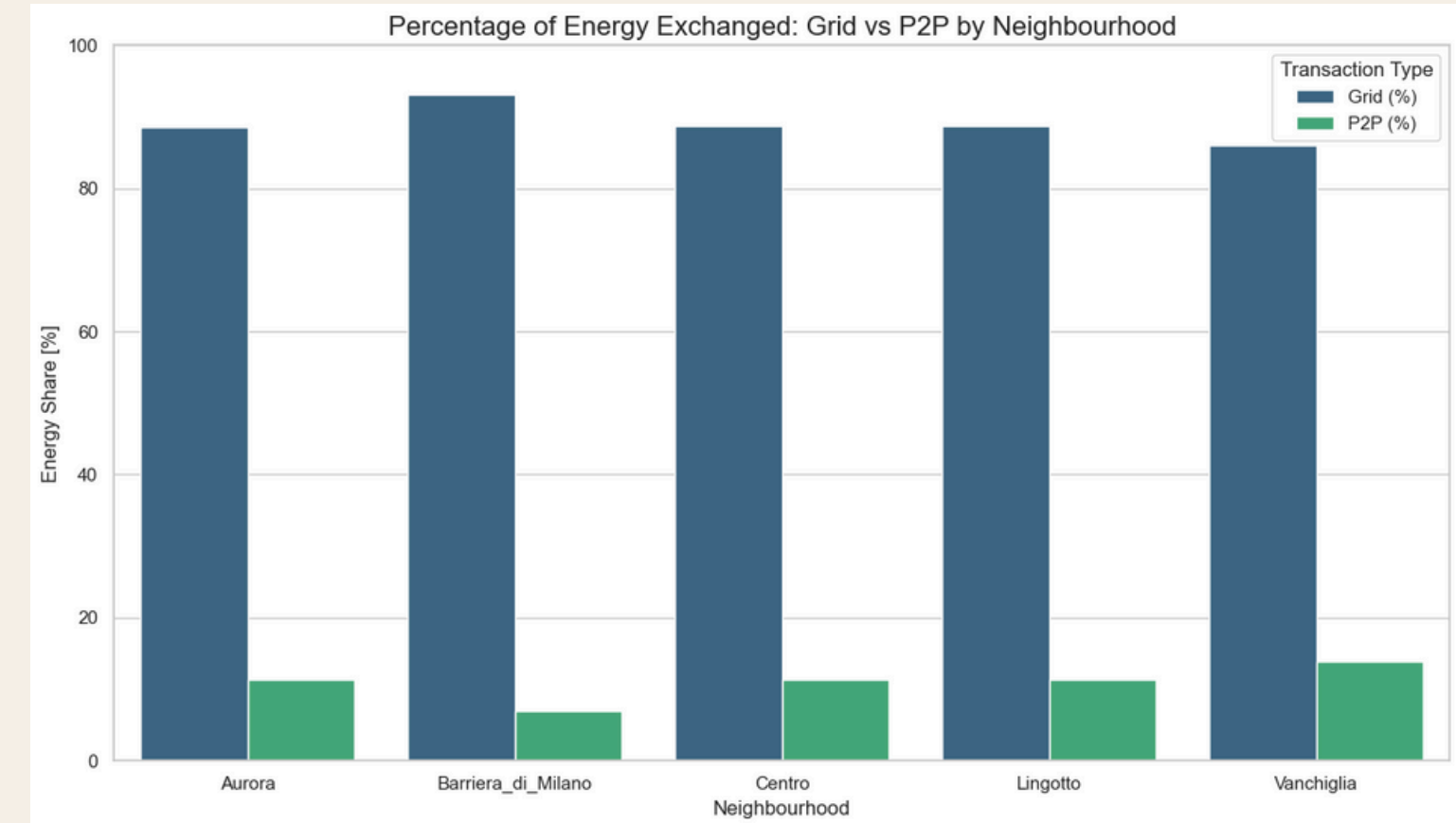


Area Chart Stacked of the Distribution of Computing Power



Conclusions

- **P2P** trading concentrated in **high PV hours** (8-16h)
→ representing targeted local energy exchange
- Overall energy flow **dominated** by **grid dependency** (85-90% of trades)
→ self-sufficiency remains challenging without aggressive regulator
- Dynamic pricing achieved **10-25% cost reduction** for **P2P** participants during **peak solar hours**
- **Transaction** throughput scaled **linearly** with **P2P** activity
→ P2P matching is efficient: 1 P2P trade replaces almost 2 grid trades (buyer + seller)



Thank you

for your attention

Project repository available [here](#)